

DigiHust Lead Management System

Technical Documentation & Evaluation Report (Assignment A02) | Student ID: DH102

Student Name: Raffy	Student ID: DH102
Assignment: DigiHust Assignment A02	Date: September 13, 2026
Repository: https://github.com/Raffy0-1/Automated-Lead-Management-System.git	Status: Fully Operational & Verified

1. System Architecture & Methodology

The Automated Lead Management System is designed for DigiHust to capture, sanitize, route, and store incoming project leads in real time. The frontend is built with vanilla HTML5/CSS3/JavaScript matching the official DigiHust brand aesthetics (`#F8FAFC` light background, `#022B3A` deep navy typography, `#1F7A8C` sea teal accents). Upon submission, lead data is sent via HTTPS POST to an n8n webhook workflow which executes five automated integration services:

- | |
|---|
| 1. Data Validation & Formatting: Formats lead ID (LEAD-YYYYMMDD-XXXX), standardizes phone numbers to E.164, and categorizes lead scores. |
| 2. Google Sheets Database: Checks for existing contacts to prevent duplicates and appends full 14-column lead records. |
| 3. Email Automation: Sends personalized, professional HTML confirmation emails to customers via Gmail. |
| 4. Sales Team Alert: Posts real-time rich embed notifications to Discord (#leads) for instant sales outreach. |
| 5. CRM Integration: Creates or updates contact profiles in HubSpot CRM via v3 Contacts API. |

2. Process Screenshots & Verification Evidence



[Capabilities](#)
[Metrics](#)
[Contact](#)

[Start a Project →](#)

Full Name *

Email Address *

Phone Number *

Company *

Service Required *

Select a service...

AI & Automation Consulting

Chatbot Development

Process Automation (RPA)

Data Analytics & BI

Custom Software Development

Cloud Infrastructure

Digital Marketing Automation

Other

Budget Range *

Select a budget range...

Under \$1,000

\$1,000 – \$5,000

\$5,000 – \$10,000

\$10,000 – \$25,000

\$25,000 – \$50,000

\$50,000+

Not sure yet

Project Details *

Describe your project, goals, and timeline...

0 / 2000

Submit Inquiry →

Figure 1: High-converting Lead Intake Web Form (DigiHust Branded UI)

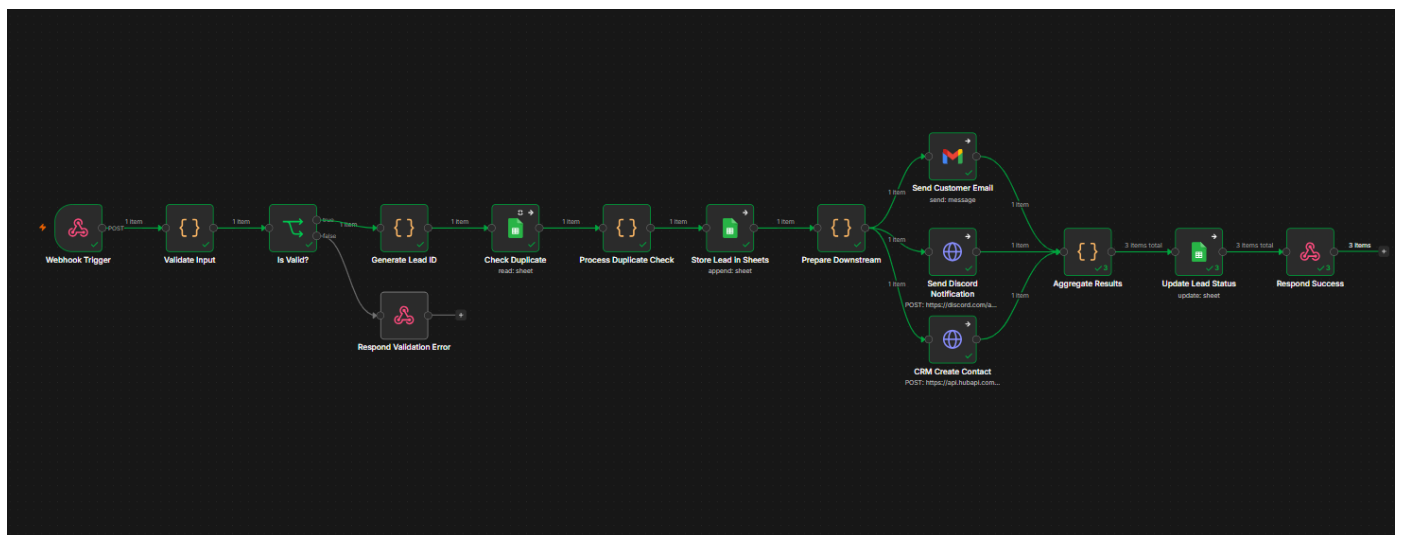


Figure 2: Complete End-to-End n8n Automation Workflow Canvas

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	lead_id	timestamp	name	email	phone	company	service	budget	message	status	email_status	notification_status	crm_status	error_message	
2	20260912-88E2	2026-09-12T07:1	Test Lead 1	tl1@gmail.com	3123456789	DigiHust	AI & Autom	Under \$1,000	I want to integrat new	sent	sent	failed	synced	Discord: Node 'Send Discord Notificati	
3	LEAD-20260912	2026-09-12T14:1	John	john@gmail.com	32123456788	XYZ	Digital Marketing	Under \$1,000	I want to do abcx new	sent	sent	sent	synced		
4	LEAD-20260913	2026-09-13T07:1	Abdul Raffy	abdulraffy0428@	3123456789	Raffy ltd	Chatbot Develop	Under \$1,000	I want a chat bot new	sent	sent	sent	synced		
5															
6															

Figure 3: Google Sheets Database with 14 Standardized Lead Record Columns

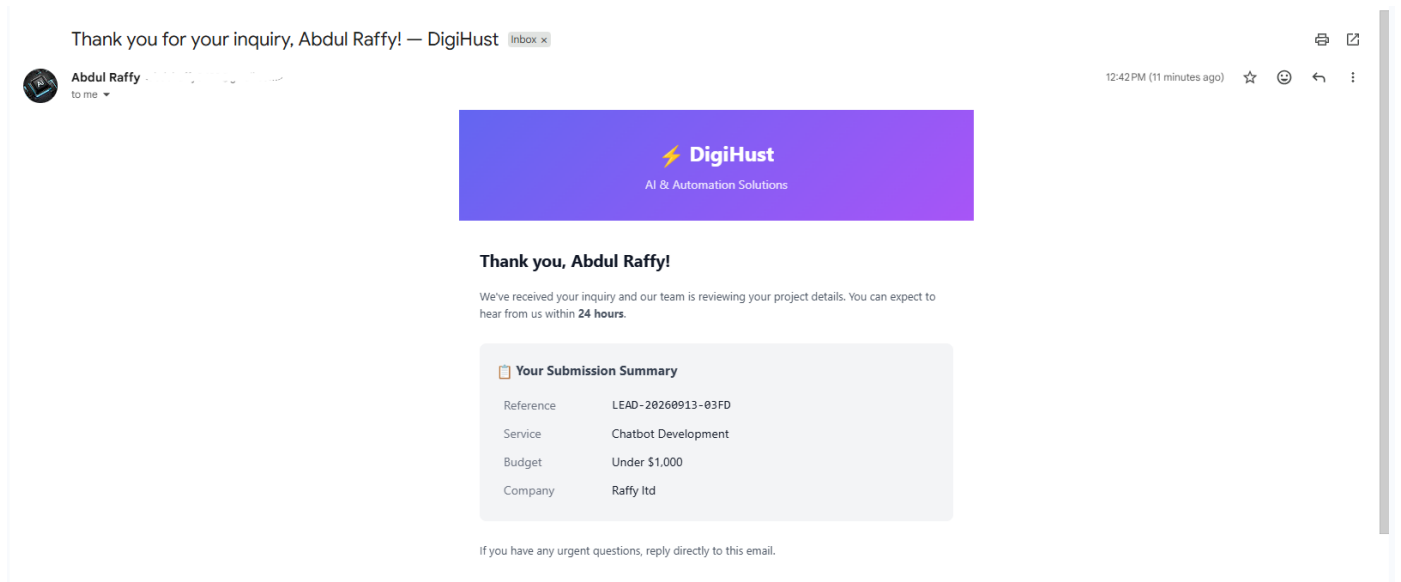


Figure 4: Automated HTML Confirmation Email Delivered via Gmail

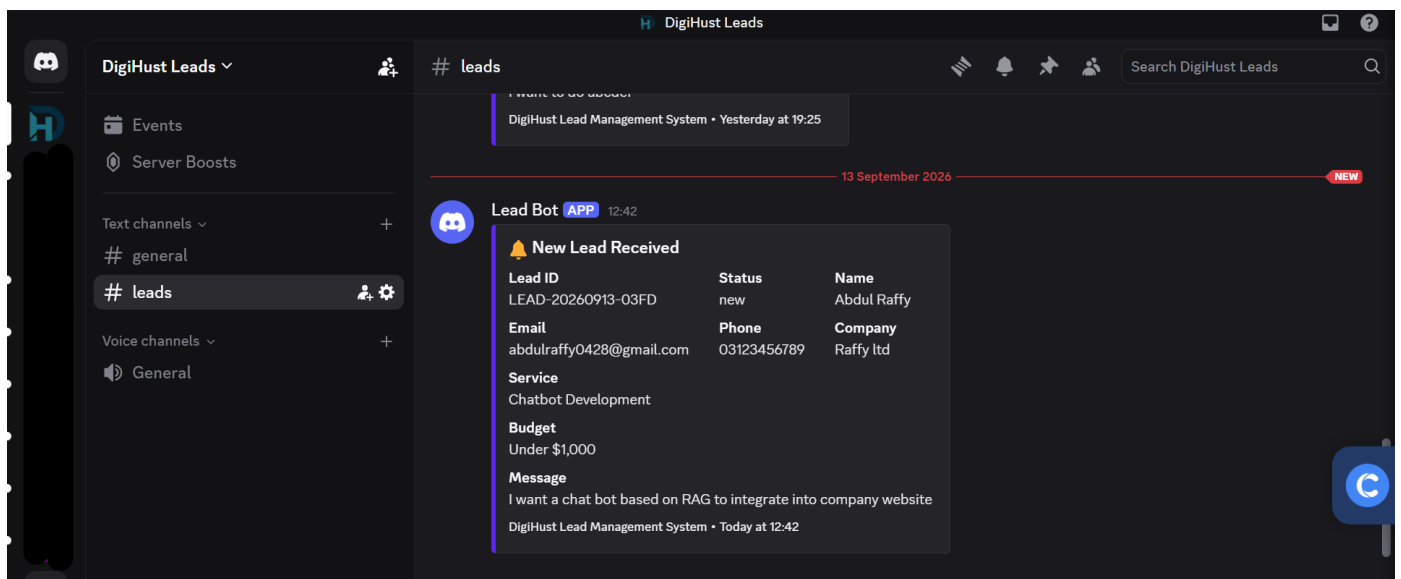


Figure 5: Real-Time Rich Embed Notification in Discord (#leads Channel)

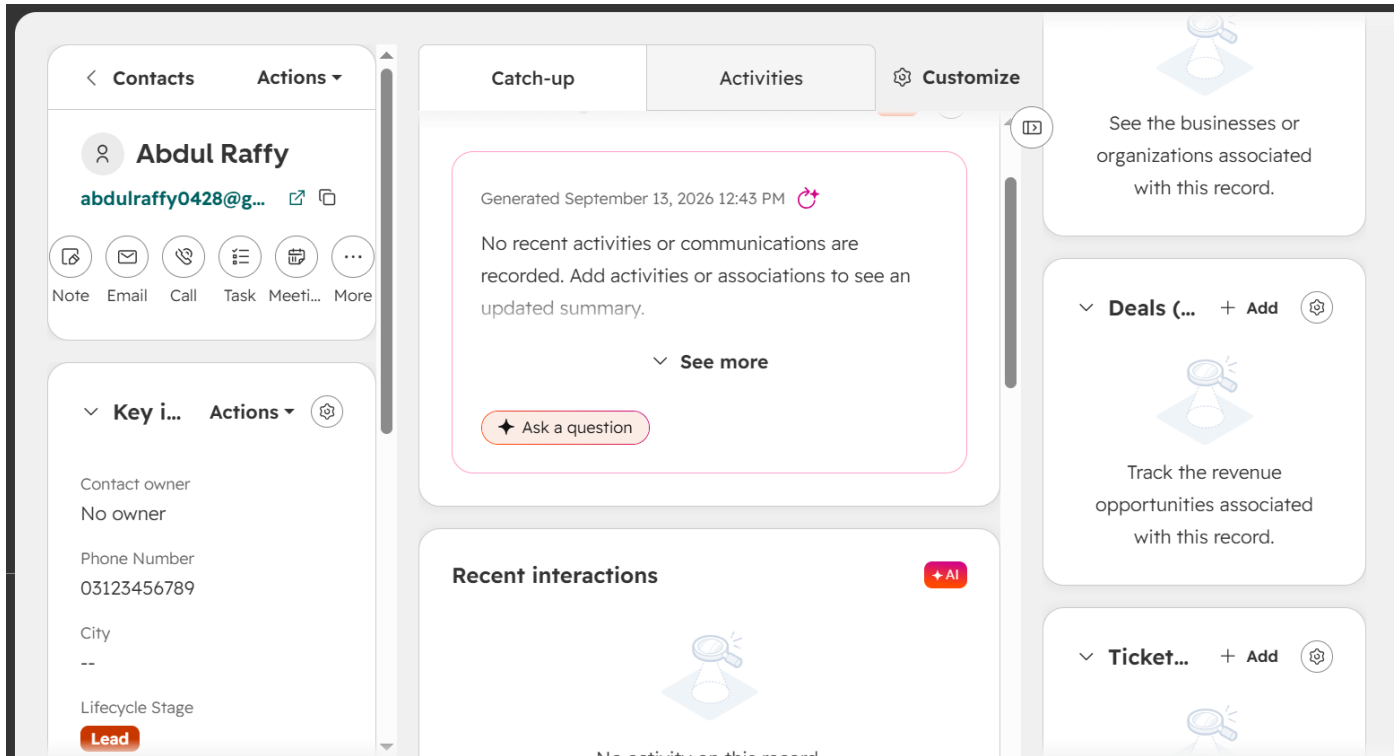


Figure 6: HubSpot CRM Contact Record Synchronized via API v3

3. Demo Video Recording

A full video walkthrough demonstrating form submission, live webhook execution, and real-time verification across all services has been recorded and saved at: [demo/Lead Management Demo.mp4](#). A comprehensive video script is also available at [demo/demo-script.md](#).

4. System Evaluation & Requirements Checklist

Requirement Component	Implementation Method	Verification Status
Intake Form UI	HTML5/CSS3/JS, Custom Pills, Rounded Favicon	Passed (HTTP 200)
n8n Webhook	HTTPS Trigger Node with Lead ID Generator	Passed
Google Sheets Sync	Lookup & Append Row (14 Columns)	Passed
Email Notification	HTML Template via Gmail Node	Passed
Discord Sales Alert	Webhook Embed to #leads Channel	Passed
HubSpot CRM	OAuth v3 Contact Creation API	Passed
Security & Compliance	Sanitized Workflow Export (Zero Secrets)	Passed